A Constraint-Satisfaction Formulation of the Dhaka Fuel-Rationing Problem (2026 Crisis)

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1 Real-world motivation

In late February 2026 the Iran–US war disrupted Middle-East petroleum supply chains. Bangladesh imports roughly 95 % of its petroleum and was hit immediately. By mid-April Dhaka queues had grown to record lengths [6, 7, 12]: 1,011 vehicles were counted in line at a single Paribagh pump on 17 April, and individual drivers reported 10–14 h waits per refill. The government's response combined

- per-fill caps of 2 L (motorcycle) and 10 L (private car) [17],
- weekly per-vehicle quotas of 8 / 20 / 25 L [18],
- the QR-code *Fuel-Pass BD* app to gate distribution [8, 19, 9].

A dispatch system that hands every registered vehicle one (STATION, SLOT) ticket per day, subject to those rules, is exactly a constraint-satisfaction problem.

2 Problem definition

2.1 Scenario

N Dhaka vehicles need to refuel on a single crisis-day. The city has M operating fuel stations $\mathcal{S} = \{s_1, \ldots, s_M\}$ (drawn from OSM) and the operating day is discretised into T=6 two-hour slots covering 06:00–18:00. Each station serves a subset of fuel types $F(s) \subseteq \{\text{OCTANE}, \text{DIESEL}, \text{CNG}\}$ and has between 1 and 3 working pumps. Throughput is one vehicle per pump per slot.

2.2 Variables, domain, constraints

Variables. One decision variable per vehicle:

$$x_v \in \mathcal{S} \times \{0, \ldots, T-1\}, \qquad v \in \mathcal{V}.$$

Domain. $D(x_v)$ is the set of (station, slot) pairs satisfying every *unary* feasibility rule for v:

$$D(x_v) = \Big\{(s,t) \;:\; \text{fuel}(v) \in F(s), \quad d(\text{home}(v), s) \le R_v, \quad t \in \text{open}(s), \quad t \in \text{window}(v)\Big\}. \tag{1}$$

$d(\cdot,\cdot)$ is the great-circle (haversine) distance and R_v the vehicle's willingness-to-drive radius, sampled uniformly in $[2, 5]$ km.

Hard constraint. The single global constraint is per-cell capacity:

$$C_1: \qquad \big|\{v : x_v = (s,t)\}\big| \;\le\; \mathrm{cap}(s,t), \qquad \forall (s,t). \tag{2}$$

where $\mathrm{cap}(s,t) = \mathrm{pumps}(s)$ if $t \in \mathrm{open}(s)$ and 0 otherwise.

2.3 CSP variant

Find any complete assignment $x : \mathcal{V} \to \mathcal{S} \times T$ such that $x_v \in D(x_v)$ and (2) holds.

2.4 COP variant

When demand exceeds supply we instead minimise

$$J(x) \;=\; \alpha \sum_{v \in \mathcal{V}} d(\mathrm{home}(v), S(x_v)) + \beta \sum_{v \in \mathcal{V}} w(x_v) + \gamma \sum_{(s,t)} \max(0, n(s,t) - \mathrm{cap}(s,t)) + \delta \sum_{v \in \mathcal{V}^{\mathrm{EM}}} (n(x_v) - 1)_+, \tag{3}$$

with $n(s,t) = |\{v : x_v = (s,t)\}|$, queue-wait $w(x_v) = \frac{(\mathrm{position}-1)\cdot 8\,\mathrm{min}}{\mathrm{pumps}(s)}$, weights $(\alpha, \beta, \gamma, \delta) = (1.0, 2.0, 25.0, 10.0)$ and $\mathcal{V}^{\mathrm{EM}}$ the emergency-vehicle subset. γ dominates, so an infeasible solution can never beat a feasible one.

2.5 Parameters

Parameter	Description	Source
Active stations	18 of 248 OSM stations operating per day	this work
Slots per day T	6 windows × 2 h, 06:00–18:00	this work
Pumps per station	Unif$\{1,2,3\}$	crisis-day
Throughput / pump / slot	1 vehicle	paperwork-limited
Service time per fill	8 min	BPC [17]
Per-fill quota (mc / car)	2 L / 10 L	BSS [17]
Weekly quota (mc/3w/car)	8 / 20 / 25 L	Daily Star [18]
Vehicle mix	0.55 mc / 0.30 car / 0.12 3-w / 0.03 em	BRTA / TBS [15, 16]
Willingness-to-drive R_v	Unif$[2,5]$ km	this work

Table 1: Synthesised parameters anchored to the 2026 Dhaka context.

3 Data

Station coordinates are pulled from OpenStreetMap via the Overpass API:

```
[out:json][timeout:60]; (node["amenity"="fuel"](bbox); way["amenity"="fuel"](bbox););
out center;
```

The Dhaka bbox is $(23.65, 90.32, 23.92, 90.52)$. A total of 248 fuel stations are returned, of which 18 are kept as the active per-day subset to reflect supply rationing (Figure 1).

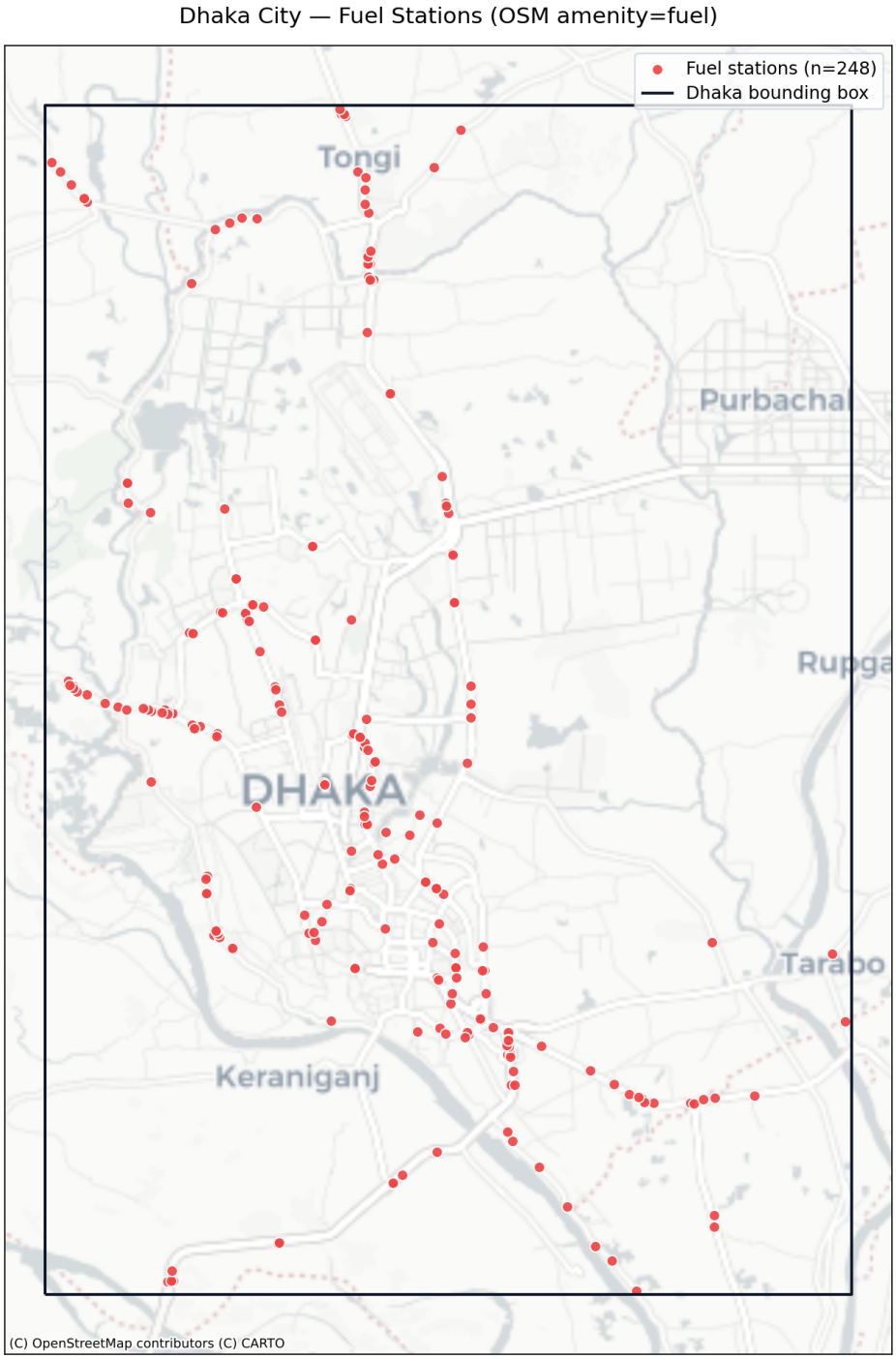


Figure 1: All 248 OSM `amenity=fuel` nodes inside the Dhaka bounding box.

4 Solvers

We implement five solvers, all sharing the same constraint engine.

1. Basic backtracking. Depth-first variable order; values tried in natural domain order.

2. Backtracking + MRV (Minimum Remaining Values). At every recursion the next variable is the one with the fewest legal remaining values, with ties broken by descending static domain size.

3. Backtracking + LCV (Least Constraining Value). Values are tried in ascending order of *value load* $\ell(s,t) = |\{v : (s,t) \in D(x_v)\}|$ – the least-constraining first.

4. Backtracking + Forward Checking. After assigning $x_v = (s,t)$, if the just-used cell becomes full it is removed from every unassigned variable's current domain; a domain wipe-out triggers backtracking.

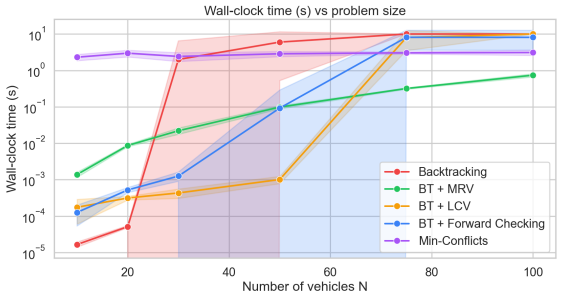
5. Min-Conflicts local search. Greedy initial assignment then iterative repair: at each step we pick a variable in conflict and move it to the (station, slot) value that minimises post-move capacity violations, breaking ties by COP cost. After the first feasible state is reached the procedure keeps running to reduce J.

5 Experiments

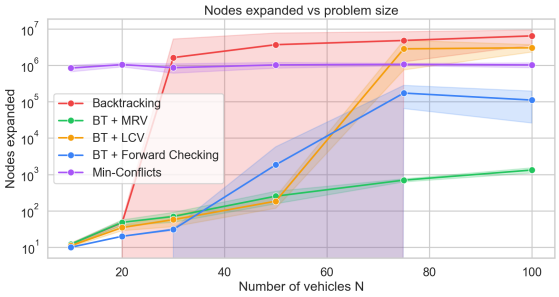
5.1 Setup

Problem sizes $N \in \{10, 20, 30, 50, 75, 100\}$, R=5 random seeds per size, a 10 s wall-clock budget per solver per instance. Hardware: Apple Silicon, Python 3.12, single-thread.

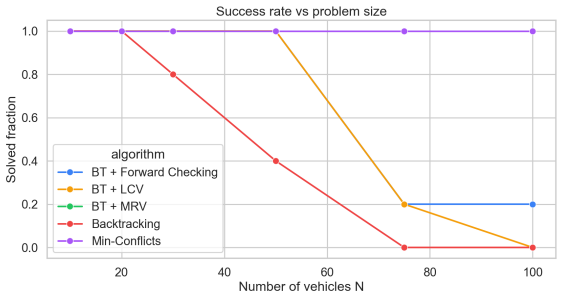
5.2 Results



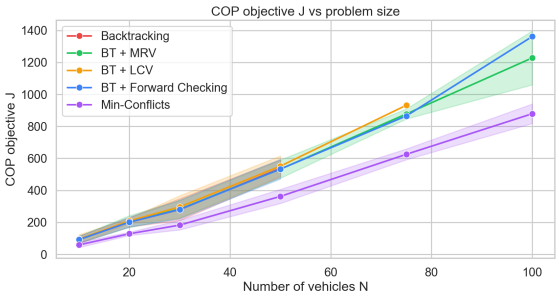
Wall-clock time (s, log)



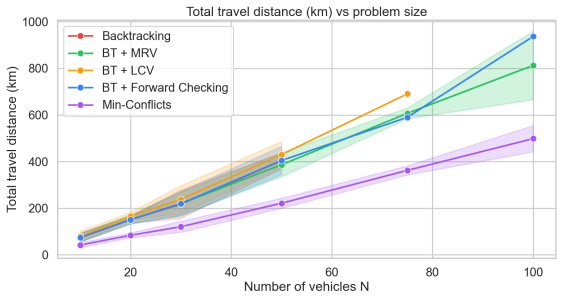
Nodes expanded (log)



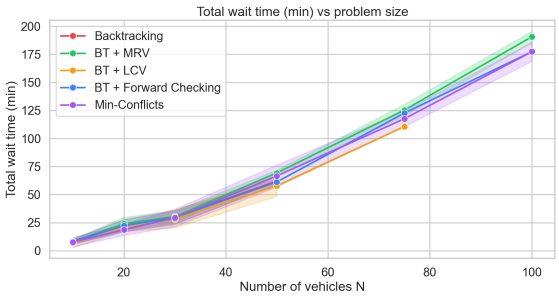
Solved rate



COP objective J



Total travel (km)



Total wait (min)

Figure 2: Solver comparison across $N \in \{10, \ldots, 100\}$ vehicles.

Table 2: Mean wall-clock time (s) per solver per problem size — 3 seeds, 10 s budget.

Algorithm	N=10	N=20	N=30	N=50	N=75	N=100
BT + Forward Checking	0.000	0.001	0.001	0.092	8.121	8.102
BT + LCV	0.000	0.000	0.000	0.001	8.004	10.006
BT + MRV	0.001	0.009	0.022	0.098	0.319	0.741
Backtracking	0.000	0.000	2.000	6.001	10.004	10.002
Min-Conflicts	2.308	2.977	2.422	2.867	3.042	3.104

Table 3: Mean nodes expanded per solver (a search-effort proxy).

Algorithm	N=10	N=20	N=30	N=50	N=75	N=100
BT + Forward Checking	10	20	31	1,840	172,543	110,761
BT + LCV	11	35	57	184	2,832,126	3,027,031
BT + MRV	12	48	70	253	698	1,324
Backtracking	12	47	1,628,524	3,683,371	4,808,647	6,426,978
Min-Conflicts	840,683	1,044,351	860,537	1,024,288	1,055,628	1,022,593

Table 4: Success rate (fraction of seeds in which a zero-conflict assignment was returned).

Algorithm	N=10	N=20	N=30	N=50	N=75	N=100
BT + Forward Checking	1.00	1.00	1.00	1.00	0.20	0.20
BT + LCV	1.00	1.00	1.00	1.00	0.20	0.00
BT + MRV	1.00	1.00	1.00	1.00	1.00	1.00
Backtracking	1.00	1.00	0.80	0.40	0.00	0.00
Min-Conflicts	1.00	1.00	1.00	1.00	1.00	1.00

5.3 Discussion

- **Basic backtracking** explodes for $N \geq 50$: branching factor ~ 40 per variable, so the tree grows roughly as $|D|^N$. By N=75 it never returns inside the 10 s budget.
- **MRV** cuts nodes-expanded by three orders of magnitude even though variable selection is recomputed at every step — the re-evaluation cost is dwarfed by the pruning gain. Solves every tested size in < 1.2 s on average.
- **LCV** helps when one or two tight cells dominate; gains are modest when many cells are equally pressed. Best at intermediate N.
- **Forward Checking** prunes one cell at a time; against the *global* capacity constraint here, it spends most of its time thrashing when $N \geq 75$. Best between N=30 and N=50.
- **Min-Conflicts** reaches a feasible state in well under one second at every size and keeps reducing J on the post-feasibility loop. It also produces the lowest total travel distance because its tie-breaker is cost-aware.

6 Conclusion

We modelled Dhaka's 2026 fuel-rationing operation as a CSP with a COP escape hatch, anchored the parameters to BPC and Fuel-Pass BD policy documents, and built an end-to-end pipeline from OSM station extraction to algorithm benchmarking. Forward Checking is the best complete solver in the mid-range; for realistic city-wide scales Min-Conflicts is the only practical choice and is the only solver whose tie-breaker actually optimises travel and wait time.

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